

Multi-domain indoor environmental quality in buildings: A review of their interaction and combined effects on occupant satisfaction

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ABSTRACT

Occupant satisfaction is influenced by the interaction and combined effects of multi-domain indoor environmental quality (IEQ) in buildings. While there are literature reviews on multi-domain IEQ, two limitations exist: 1) the reviewed IEQ interactions are mainly limited to two IEQ domains and climate chamber studies; and 2) those reviews have not quantitatively discussed the impact of IEQ interactions on overall satisfaction and only cover limited building types. To address this gap, this study adopts a broad, top-down approach with a three-stage screening to select relevant publications about the four most concerned IEQ domains—thermal comfort, acoustic comfort, visual comfort, and indoor air quality. A total of 74 publications have been selected and reviewed. Our study shows that: 1) existing findings on IEQ interaction and combined effects are inconsistent and somewhat contradictory; 2) interactions among three IEQ domains exist but are much less examined compared to two IEQ interactions. Studies on four IEQ are not found; 3) the current occupant satisfaction models are mainly generalized models. Personal models are an emerging area of research and practice; and 4) the one-vote veto effect exists, i.e., some IEQ domains show a predominating influence on overall satisfaction. We advocate future research 1) to collaborate on a global IEQ database to enable model comparison and the development of standards and guidelines spanning multiple IEQ domains, 2) to investigate personal overall satisfaction models encompassing IEQ interactions; and 3) to further examine non-linear modeling approaches for interpreting the relationships between IEQ and overall satisfaction.

1. Introduction

The indoor environmental quality (IEQ) affects occupants' perception and satisfaction [1,2], health and well-being [3,4], work productivity and performance [4–7], and organizational effectiveness [8]. Improved IEQ conditions can bring about a range of social, economic, and environmental benefits. The U.S. Green Building Council has estimated that improved indoor conditions could result in worker performance increases worth \$90 billion by 2030 [9]. Adequate IEQ can prevent the sick building syndrome (SBS) symptoms, such as lethargy, headaches, and fatigue [3,5], and induce positive comprehensions and moods [6]. Such health impacts are particularly important for the most vulnerable population to the adverse effects of poor IEQ environments, such as the very young, seniors, and people with cardiovascular or respiratory diseases who often stay longer indoors [10].

On the other hand, IEQ conditions significantly affect the building energy footprint and emissions, as more than 80% of building energy is consumed during the occupancy phase [11], in which occupants interact

with indoor environments and systems in the daily routine to meet their needs and requirements through various behaviors (e.g., adjust thermostats, turn on the lights). However, lack of comfort is still a common issue as reported by many literature, despite the enormous amount of energy used to control the environments (e.g. [12]).

IEQ is a multi-domain variable which includes all aspects of an indoor environment that relate to the comfort, performance, and health of its occupants, including thermal comfort and control, indoor air quality (IAQ), lighting quality, daylight and exterior views, acoustics, noise and vibration control, odors, electromagnetic radiation, potable water monitoring, ergonomics, among other factors [13]. As occupants live in a complex built environment, their IEQ perceptions are influenced by a multitude of environmental and human factors, each affecting one or more IEQ domains [14–16]. Meanwhile, different IEQ domains collectively affect an occupant's overall satisfaction. **Such combined effect on overall satisfaction is not a simple superposition of individual IEQ but should be evaluated in an integrated way by including the impact of IEQ interactions.** For instance, Geng et al. [17] observed that

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occupants raised their expectations of other IEQ domains when they were thermally satisfied, which correspondingly changed their overall satisfaction. Fig. 1 shows the interconnections between the three aforementioned levels—overall satisfaction, multi-domain IEQ, and environmental and human factors. In this figure, there are two types of effects: 1) *the combined effect*, which refers to the aggregated impact of multiple factors at a lower level on a single factor at a higher level. For example, the effect of multi-domain IEQ on overall satisfaction (i.e., level 2 to level 1) can be determined by weighing the contribution of each IEQ to overall satisfaction. Similarly, the multi-faceted environmental and human factors (level 3) have a combined effect on individual IEQ perception (level 2); and 2) *the interaction effect*, which is the interplay of two or more factors at the same level [21]. For example, thermal comfort and IAQ, both are the domains of IEQ, interact with each other and collectively affect overall satisfaction—when occupants feel warm, they perceive the air quality to be worse even when its condition remains unchanged [19].

While there exist literature reviews on multi-domain IEQ, their limitations are twofold: 1) for the IEQ interactions, existing reviews are mainly limited to two IEQ domains [20,21] or those conducted in climate chambers [20]. Interactions involving three and more IEQ and field evaluations have not been sufficiently discussed; and 2) for the combined IEQ effects on overall satisfaction, those reviews have not quantitatively discussed how IEQ interactions affect overall satisfaction (e.g., how should different interactions be weighed?) and only cover limited building types [21]. For example, Torresin et al. [20] examined the climate chamber studies published between 1990 and 2017 which primarily focused on the interactions of two IEQ and their respective environmental factors on occupants' perception and performance in non-industrial buildings (e.g., offices, schools). This study concluded a few challenges in multi-domain IEQ research, such as the complexities of full factorial experiment design and difficulties in selecting indicators of each IEQ. It also highlighted the limitations in current research design (e.g., limited control variables, small sample size, exclusion of extreme conditions) and in performance assessment (e.g., neglecting occupants' learning ability and fatigue). Since the publication of this review, there have been several new studies on this topic (e.g., technical studies that included interactions when calculating overall satisfaction [23–25,79], studies on three IEQ interactions [18,27–30]), all providing new evidence of the underlying interaction and combined effects. Therefore, this study aims to fill the knowledge gap in literature review by 1) examining the new findings on combined effects of IEQ on overall

satisfaction, with emphasis on different IEQ interactions involved, modeling approaches, and comparison of IEQ importance; and 2) reviewing recent studies in both climate chambers and field evaluations of various building types.

This review considers studies that address two or more IEQ domains, those that only examine a single domain are excluded. In particular, we focus on four IEQ domains—thermal comfort, acoustic comfort, visual comfort, and IAQ, given their prevalence in occupant satisfaction studies over the last two decades. This review is organized as follows: Section 2 introduces the influential factors of IEQ and occupant satisfaction. Section 3 presents the review methodology. Section 4 summarizes the approaches to collect occupant feedback and environmental data. Section 5 discusses the combined effects of IEQ on overall satisfaction. Section 6 discusses the IEQ interactions. Section 7 presents our review findings. Finally, section 8 concludes this study.

2. Background

Besides the common environmental factors (e.g., air temperature, illuminance, sound pressure level), this section summarizes four frequently discussed influential factors which affect IEQ perceptions and occupants' satisfaction.

2.1. Building layout

Building layout, including the floor plan, locations of equipment, aesthetics, and furniture, has a great impact on occupant satisfaction [31–33]. Choi et al. [34] argued that radiant asymmetry should be considered in architectural designs as occupants working in perimeter zones were more affected by the outdoor climate than those working in interior zones. Yildirim et al. [35] showed that workplace satisfaction in an open-plan office was negatively correlated with occupants' proximity to windows. Likewise, different seating plans (e.g., window seat, door seat) would lead to varying perceptions of ventilation, air freshness, natural lighting, and noise in offices [32]. ASHRAE-55 [36] suggests the temperature difference between the inside surface of a window and the interior zone should be less than 10 °C to maintain thermal comfort. Moreover, having access to nature scenes has shown long-term restorative effects and can improve physical and psychological well-being [37]. This concept is generally referred to as biophilia [38], which connects occupants with nature and other living beings by increasing their connectivity to natural environments, such as plants, water, and

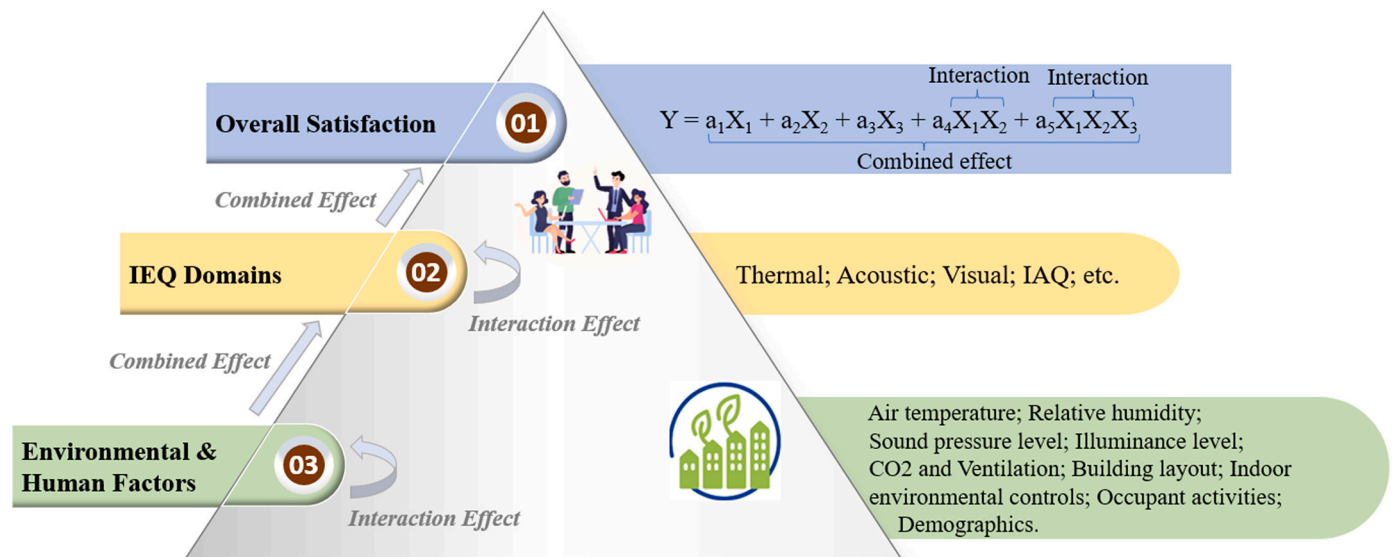


Fig. 1. Interconnections between overall satisfaction, IEQ domains, and environmental and human factors.

analogue biomorphic forms [5]. The principles of biophilia are widely adopted in building designs, especially in the workplace, and have shown multiple benefits, such as improved IAQ and health [39].

2.2. Indoor environmental controls

Occupants' accessibility to indoor environmental controls (IECs) (e.g., thermostat, window, lighting) is directly associated with their IEQ perceptions and reflects their actual controls over built environments [14]. IECs play a key role in satisfying personal environmental preferences [40], reducing energy consumption without sacrificing comfort [41,42], facilitating productivity [43,44], and preventing health-related issues [45].

Lack of IECs has been a common issue in commercial buildings. A study of 167 recently built or refurbished offices found that the majority of participants (63%) declared no or little control over the sound, followed by ventilation (53%) and temperature (47%) [44]. The low IECs can result from the lack of physical control (occupants are not allowed or unable to do so) or lack of control knowledge (occupants do not know how to do it) [44,46]. The accessibility of IECs was showed to be negatively associated with the occupancy level, i.e., occupants have fewer IECs in shared open-plan offices than in low-occupancy private offices [14,44].

2.3. Occupant activities

Occupant activities can affect the IEQ perceptions. For example, to control the impacts of activities on thermal comfort, researchers usually prescribe the activities performed in climate chamber experiments (e.g., sitting: 1.0 Met, web browsing: 1.2 Met) [25,47]. Similarly, occupants in design work were more disturbed by machine noises than those in programming or reading [32].

2.4. Demographics

Gender, culture, age, and social influence can result in different IEQ perceptions. Choi et al. [48] noted that females were less satisfied than males when the temperature was 23–24 °C due to the varying clothing insulations and metabolic rates [1,48]. Kang et al. [32] observed that females had higher requirements for ventilation and personal spaces. Culture is another influencing factor [14]. People from different birthplaces and cultural backgrounds had different IEQ preferences, habits (e.g., clothing), and adaptive behaviors (e.g., interactions with IECs). Slovenians were more outspoken about thermostatic control, lighting, and shades than Italians [14]. In open-plan offices, older occupants (≥ 24 years) tended to be more sensitive to lighting and noise than younger counterparts [32]. Moreover, social influence, such as group norms and conformity intention [49], also plays a role in affecting an occupant's behaviors and IEQ perceptions [14].

3. Methodology

A broad, top-down approach is adopted to select initial relevant publications using broad terms and then to exclude those that are outside the scope [50]. The preliminary selection process consists of four steps: 1) create the query of keywords and the logic grid; 2) identify academic databases, source types, and publication years; 3) conduct screening; and 4) evaluate and synthesize the relevant publications. This section focuses on the first three steps, and the results from the last step are presented in Sections 5–6.

3.1. Create the query of keywords and the logic grid

Creating a comprehensive list of keywords is essential for retrieving all relevant publications. As stated in the introduction, this review focuses on the four major IEQ domains, i.e., thermal comfort, acoustic

comfort, visual comfort, and IAQ. Since the primary objective is to examine the effect of multi-domain IEQ on occupants' overall satisfaction, the following keywords are selected, including "buildings", "building environments", "built environments", "occupants", "humans", "inhabitants", "residents", "overall experience", "overall comfort", "overall perception", "overall satisfaction", "overall evaluation", "overall behavior", "indoor environmental quality (IEQ)", "IEQ comfort", "IEQ evaluation", "environmental factors", "thermal", "acoustic", "visual", "indoor air quality", "temperature", "sound", "lighting", "air", "interaction effect", "cross effect", "combined effect", "comprehensive effect". A logic grid of keywords is created correspondingly (see Table 1).

3.2. Database, source types, and publication years

Three academic databases—Google Scholar, Web of Science, and ScienceDirect—are selected since they provide good coverage of publications in building science, engineering, architecture, and social and behavioral science. The source types include peer-reviewed journals, conference proceedings, and book chapters written in English. Industry reports, websites, magazines, and dissertations are excluded. The publication date is set to return publications after the year 2000. After delimiting the scope, 640 papers are initially identified in Google Scholar, 394 in Web of Science, and 288 in ScienceDirect as of March 2022. The duplicates across three databases are only counted once.

3.3. Screening process

A three-stage screening (see Fig. 2) is conducted to filter the publications according to their keywords and abstracts. In the first stage, only publications that cover two or more focal IEQ domains in indoor environments are retained, resulting in 154 papers in total. In the second stage, publications that meet the following criteria are further selected: 1) present details about the experimental design; and 2) consider IEQ interactions or combined effects of IEQ on overall satisfaction. 52 publications are identified after the screening. In the third stage, an in-depth screening of the reference list of each publication identified in stage 2 is conducted to pick up additional relevant publications that have not yet been included. For those publications, the publication year is not restricted. 22 additional publications are collected as a result. Thus, a total of 74 publications are reviewed in detail.

4. Methods to evaluate IEQ and overall satisfaction

Understanding how occupants' satisfaction and environmental conditions are measured is a prerequisite to interpret the results of interaction and combined effects. Post-occupancy evaluation (POE) is the most common means to capture occupants' IEQ satisfaction during the occupancy phase. It also plays a significant role in measuring buildings' actual performance (e.g., comfort, energy consumption) and thus is often used in building rating or certification systems to validate if the original design and operational objectives have been met [4,51]. Common POE approaches include occupant survey and environment

Table 1
Keywords and the logic grid.

Category	Keyword
Environment	"buildings" OR "built environments" OR "building environments"
People	"occupants" OR "humans" OR "inhabitants" OR "residents"
Experience	"overall experience" OR "overall comfort" OR "overall perception" OR "overall satisfaction" OR "overall evaluation" OR "overall behavior" OR "IEQ comfort" OR "IEQ evaluation"
IEQ	"thermal" OR "visual" OR "acoustic" OR "indoor air quality" OR "temperature" OR "sound" OR "lighting" OR "air"
Effect 1	"interaction effect" OR "cross effect"
Effect 2	"combined effect" OR "comprehensive effect"

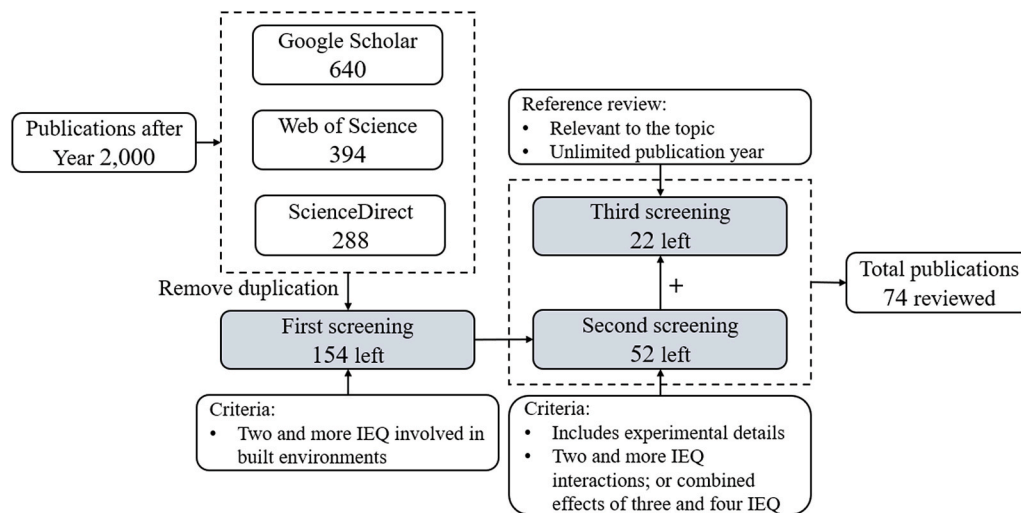


Fig. 2. Method to identify relevant publications.

monitoring, which are described in the subsections below.

4.1. Occupant survey

Paper-based or digital surveys are frequently used in occupant feedback collection. Occupant surveys are useful tools to study IEQ satisfaction and building performance. Well-known examples include the Center for the Built Environment (CBE) survey and the Building Use Studies (BUS) survey. The former collected 52,980 responses from approximately 900 buildings [12] and the latter surveyed over 700 buildings [51]. The way survey questions are phrased has a considerable impact on respondents' interpretation and how researchers derive the conclusions [52]. Occupant surveys use a suite of Likert scales to benchmark IEQ perceptions and comfort, including the ASHRAE seven-point votes [17,22,24,28,40,53], three-/five-/eleven-/thirteen-point comfort votes [30,47,54–57], 0–100 analogue scale comfort votes [58–60], and 10-point acoustic, visual, and IAQ comfort votes [25]. Occupants may be asked to take the survey more than once or repeat it at certain intervals, especially in longitudinal studies (e.g., [17,22,24,57,61]). However, a comprehensive occupant survey is often lengthy and time-consuming [54], which can lead to decreased response rates or survey fatigue over time [62,63].

4.2. Environment monitoring

Currently, there is no standard data collection tool for environment monitoring [52,64]. Researchers usually adopt the building automation system (BAS) or networked wired or wireless sensors to collect environmental data. Building standards (e.g., WELL Building Standard) suggest that at least one sensor is needed for every 300 m². However, a higher sensing density (76 m² per sensor) is desired to get a finer IEQ resolution [63]. Besides the consideration of sensing density, the direct exposure of sensors to environmental stimuli (e.g., direct sunlight, air path, CO₂ breathed out by occupants) should be avoided when determining their deployment locations [52]. Moreover, other data collection means including stationary sensor kits [62,65], mobile IEQ carts [34, 66], hand-held IEQ sensors [67] have also been applied in existing research. Nevertheless, these methods require demanding calibration or maintenance efforts [60] and need to be manually relocated if the focal locations change.

5. The combined effects of IEQ on overall satisfaction

Humphreys [68] indicated that satisfaction/dissatisfaction with a

single IEQ domain did not imply satisfaction/dissatisfaction with the overall environment. However, this statement appeared to be somewhat contradictory to other studies (e.g. [55,69]), which argued that an unsatisfying IEQ domain had a significant negative impact on overall satisfaction and can hardly be compensated by a higher satisfaction with other domains. These viewpoints emphasized that occupants' overall satisfaction is not a causal, reducible relationship but is an actual lived experience [70], which is manipulated by various IEQ domains and would change over time. Studies have reported that indoor environments following building IEQ standards and guidelines did not guarantee occupant satisfaction [71]. This is mainly because most such standards and guidelines address IEQ domains separately rather than in an integrated manner [72], along with other explanations such as different building types and populations, and personal variations [73].

In the following sections, we summarized the state-of-the-art findings of how multi-domain IEQ affects overall satisfaction based on 19 relevant studies, and the model details were presented in Table 2. Most of the selected studies were carried out in real-world buildings, except [25], which was completed in a climate chamber. To better understand the relationships of three levels illustrated in Fig. 1, we classified the studies by their aggregation approaches: 1) direct correlations between environmental factors and overall satisfaction (level 3 to level 1; Section 5.1); and 2) correlations between environmental factors and overall satisfaction through IEQ domains as mediators (level 3 to level 2 and then to level 1; Section 5.2). Besides, the one-vote veto effect reported by some studies was also discussed (Section 5.3). After that, the summary of combined effects was presented (Section 5.4).

5.1. Correlations between environmental factors and overall satisfaction

Humphreys [68] directly modeled the impact of occupants' satisfaction with environmental factors (temperature, air movement, humidity, illuminance, sound pressure level (SPL), air quality) on their overall satisfaction using survey data from 26 offices in five European countries (France, Greece, Portugal, Sweden, UK). The results showed that satisfaction with temperature and air quality ranked high in terms of their impacts on overall satisfaction, while satisfaction with illuminance ranked low. However, due to different cultures and environment preferences, the weights estimated from the data were different across countries, especially for air movement—it was a significant factor in France but not in Greece, making the development of international standards for IEQ impacts a challenging task.

Table 2
Summary of overall satisfaction models.¹

Study	Experiment design	Participant	Environmental data	Modeling approach	Prediction function	Overall satisfaction
[68]	Office (field)	4,655 occupants	T, globe T, RH, V-air, SPL, I, CO2	Multivariate linear regression	$S = 1.24 + 0.39 \times S(\text{temperature}) + 0.16 \times S(\text{air movement}) + 0.12 \times S(\text{humidity}) + 0.05 \times S(\text{illuminance}) + 0.13 \times S(\text{sound}) + 0.36 \times S(\text{air quality}), R^2 = 0.540$	N/A
[64]	Office (field)	422 occupants	T, radiant T, globe T, SPL, I, CO2	Multivariate linear regression	$PDIEQ = 0.42 \times PDTC + 0.09 \times PDIAQ + 0.28 \times PDAC$ $PDTC = (0.021 \times T^2 - 0.919 \times T + 10.225) \times 100\%$ $PDIAQ = [0.28 \ln(\text{CO}_2) - 1.68] \times 100\%$ $PDAC = [0.0389 \times \text{SPL} - 1.9652] \times 100\%$	Level 1: 10% dissatisfaction (better than average grade) Level 2: 20% dissatisfaction (an average grade) Level 3: 30% dissatisfaction (below average grade)
[74]	Office (field)	68 occupants	N/A	Multivariate linear regression	$S = 0.30 \times ST + 0.36 \times SIAQ + 0.16 \times SV + 0.18 \times SA, R^2 = 0.940$ $ST = 95 \times e^{-0.03353 \times PMV^4 - 0.2179 \times PMV^2}$ $SA = 100 - 2 \times (\text{Actual SPL} - \text{Design SPL})$ $SV = -176.16 X^2 + 738.4X - 690.29$, where $X = \{\ln(\ln(I))\}$ $SIAQ = 100 - PDIAQ$	N/A
[55]	Office, educational building, library (field)	500 occupants (292 M, 208 F) Age: 20 to 30 Ci: 0.97 clo	T: 16.6–30.3 °C RH: 15%–75% V-air: 0.01–0.44 m/s I: 140 to 2,150 lx SPL: 39–56 dB CO2: 275 to 2,360 ppm	Multivariate linear regression	$S = 0.075 + 0.316 \times ST + 0.118 \times SIAQ + 0.171 \times SV + 0.224 \times SA, R^2 = 0.465$ $ST = -0.0063 \times T^2 + 0.287 \times T - 2.934$ $SA = -0.0230 \times \text{SPL} + 1.382$ $SV = -5 \times 10^{-7} \times I^2 + 0.0011 \times I - 0.106$ $SIAQ = -0.0002 \times \text{CO}_2 + 0.244$	$S > 0$: satisfied $S < 0$: dissatisfied
[58]	Educational building (field)	115 students and faculty	T, SPL, I	Self-reported IEQ weights, linear model	$S = (0.251 \times ST + 0.241 \times SA + 0.244 \times SV + 0.263 \times SIAQ) / 0.999$ $ST = 28.57 \times T - 514 (T \leq 21.5 \text{ °C}) \text{ or } -28.57 \times T + 800 (T \geq 21.5 \text{ °C})$ $SA = -0.33 \times \text{SPL} + 200$ $SV = 0.33 \times I$	Class A: $S > 90$ Class B: $90 \geq S > 67.5$ Class C: $67.5 \geq S > 32.5$ Class D: $32.5 \geq S > 10$ Class E: $10 \geq S$ $S > 0.5$: satisfied $S \leq 0.5$: dissatisfied
[54]	Classroom (field)	928 students	Mean values: T: 24.13 °C RH: 47.14% V-air: 0.11 m/s SPL: 41.69 dB I: 251.29 lx	Self-reported IEQ weights, linear model	$S = 0.35 \times ST + 0.35 \times SA + 0.3 \times SV$ $ST = (PMV + 1) - (PMV \times \frac{4}{3}) $ $SA = (0.3 \times \text{IBN} + 0.3 \times \text{IEN} + 0.2 \times \text{ISQ} + 0.1 \times \text{IVP} + 0.1 \times \text{ILQ}) / 10$ $SV = (0.2 \times \text{ALQ} + 0.1 \times \text{ALS} + 0.2 \times (10 - \text{ALG}) + 0.3 \times (10 - \text{NLR}) + 0.2 \times \text{NLQ}) / 10$	$S > 0.5$: satisfied $S \leq 0.5$: dissatisfied
[23]	Hospital(field)	875 occupants	T, RH, SPL, sound privacy, I, odor	Multivariate linear regression using structural equation modeling	$S \sim 0.83 \times \text{IEQ(PM)}, R^2 = 0.700$ $\text{IEQ(PM)} \sim 0.62 \times \text{QT} + 0.50 \times \text{QA} + 0.48 \times \text{QV} + 0.42 \times \text{QIAQ}$ $\text{QT} \sim 0.81 \times S(T) + 0.81 \times S(RH) + 0.84 \times S(\text{thermal})$ $\text{QA} \sim 0.78 \times S(\text{SPL}) + 0.78 \times S(\text{sound privacy}) + 0.84 \times S(\text{acoustic})$ $\text{QV} \sim 0.81 \times S(I) + 0.88 \times S(\text{visual})$ $\text{QIAQ} \sim 0.82 \times S(\text{odor}) + 0.90 \times S(\text{IAQ})$	Variances in satisfaction accounted by IEQ(PM): 90%: IEQ is above average performance 80%: IEQ meets average performance 70%: IEQ is below average <70%: IEQ is poor
[75]	Office (field)	293 occupants	T: 18–25 °C SPL: 45–72 dB I: 200 to 1,600 lx CO2: 500 to 1,800 ppm	Multivariate logistic regression	$S = 1 - (1 + e^{15.02 + 6.09 \times ST + 4.88 \times SIAQ + 3.7 \times SV + 4.74 \times SA})^{-1}$ $ST = 1 - \text{PPD}/100$ $SA = 1 - (1 + e^{9.540 - 0.134 \times \text{SPL}})^{-1}$ $SV = 1 - (1 + e^{-1.017 + 0.00558 \times I})^{-1}$ $SIAQ = 1 - \{(1 + e^{3.118 - 0.00215 \times \text{CO}_2})^{-1} - (1 + e^{3.230 - 0.00117 \times \text{CO}_2})^{-1}\} / 2$	$S > 0.5$: satisfied $S \leq 0.5$: dissatisfied
[76]	Residential building (field)	125 occupants: (43 M, 82 F)	T: 23–31.5 °C radiant T: 22.5–31.3 °C operative T: 22.8–31.4 °C	Multivariate logistic regression	$S = 1 - (1 + e^{-33.63 + 22.05 \times ST + 1.609 \times SIAQ + 11.77 \times SV + 21.86 \times SA})^{-1}$ $ST = 0.95 e^{-0.2179 \times (2.2 \text{PMV} - 0.15)^2 - 0.03353 \times (2.2 \text{PMV} - 0.15)^4}$	$S > 0.5$: satisfied $S \leq 0.5$: dissatisfied

(continued on next page)

Table 2 (continued)

Study	Experiment design	Participant	Environmental data	Modeling approach	Prediction function	Overall satisfaction
		Age: 32 Ci: 0.1–0.67 clo	RH: 67%–100% V-air: 0.08–0.75 m/s I: 187 to 1,522 lx SPL: 67–78 dB CO2: 330 to 1,500 ppm		$SA = 1 - (1 + e^{23.75 - 0.2986 \times SPL})^{-1}$ $SV = 1 - (1 + e^{-14.13 + 0.9058 \times I})^{-1}$ $SIAQ = 1 - (1 + e^{45.24 - 0.0257 \times CO2})^{-1}$	
[25]	Office (climate chamber)	8 students: (4 M, 4F) Age: 23 ± 1 Ci: 0.5 clo	PMV: −1.5 to 1.5 V-air: <0.2 m/s SPL: 40–65 dB I: 100 to 1,000 lx color temperature: 4,000 K CO2: 500 to 1,500 ppm	Multivariate linear regression, arithmetic mean model, geometric mean model	Multivariate linear regression: $S = -2.384 + 0.463 \times ST + 0.529 \times SA + 0.423 \times SV - 0.136 \times SIAQ$, $R^2 = 0.911$ $S = -3.106 + 0.438 \times ST + 0.401 \times SV + 0.525 \times SA$ (without IAQ), $R^2 = 0.908$ Arithmetic mean model: $S = -3.282 + 1.381 \times (ST + SV + SA)/3$, $R^2 = 0.893$ Geometric mean model: $S = -2.656 + 1.306 \times (ST \times SV \times SA)^{1/3}$ For each IEQ: $ST = 7.88 + 0.254 \times PMV + 1.51 \times PMV^2$ $SA = 16.8 - 0.203 \times SPL - 2.92 \times 10^{-3} \times I + 4.66 \times 10^{-5} \times (SPL \times I)$ $SV = 7.80 - 4.87 \times (1 + e^{(I-210)/86.2})^{-1}$ $SIAQ = 8.97 - 0.326 \times e^{1.10 \times PMV - 0.011 \times SPL}$	N/A
[61]	Classroom (field)	224 students and staff: (135 M, 89F)	T: 19–30 °C RH: 48.5%–72.6% V-air: 0.36–0.53 m/s I: 239–919 lx SPL: 46.6–60.2 dB CO2: 449 to 1,641 ppm	FCE, AHP	$S = 0.3177 \times ST + 0.1629 \times SIAQ + 0.2386 \times SV + 0.2808 \times SA$ $ST = -0.0117 \times T^2 + 0.5979 \times T - 6.878$ $SA = -0.01216 \times SPL + 1.452$ $SV = -4.838 \times 10^{-7} \times I^2 + 8.179 \times 10^{-4} \times I + 0.4833$ $SIAQ = -0.0004212 \times CO2 + 0.9756$	N/A
[77]	Office (field)	1,352 occupants	T: 26.66 °C (summer) 25.50 °C (winter) 23.82 °C (transition seasons) RH: 52.84% (summer) 18.71% (winter) 29.23% (transition seasons) SPL: 39.82–60.65 dB I: 38.50 to 1,600.42 lx CO2: 492.69 to 1,450.97 ppm PM: 3.12 to 82.75 µg/ m ³ globe T, V-air	FAHP, entropy method	$S = 0.318 \times ST + 0.301 \times SA + 0.157 \times SV + (0.102 \times S(\text{air freshness}) + 0.122 \times S(\text{air pollution}))$ $ST = -0.757 \times PMV^2 - 0.240 \times PMV + 5.573$ for summer $= -0.457 \times PMV^2 - 0.475 \times PMV + 4.979$ for winter $= -0.435 \times PMV^2 - 0.350 \times PMV + 5.638$ for transition seasons $SA = -0.117 \times SPL + 10.053$ $SV = -2.05 \times 10^{-6} \times L^2 + 3.94 \times 10^{-3} \times L + 3.410$ $S(\text{air freshness}) = 7.96 \times (CO2 - 396.93)^{-0.11}$ $S(\text{air pollution}) = 7.88 \times (PM2.5 + 11.28)^{-0.18}$	N/A

¹ 1. M: male, F: female, Ci: clothing insulation.

2. Methods: FCE = fuzzy comprehensive evaluation, AHP = analytic hierarchy process, FAHP = fuzzy analytic hierarchy process.

3. IEQ measurement: S = overall satisfaction, ST = thermal comfort, SA = acoustic comfort, SV = visual comfort, SIAQ = IAQ comfort, PDIEQ = percentage of dissatisfaction in overall satisfaction, PDTC = percentage of dissatisfaction in thermal comfort, PDAC = percentage of dissatisfaction in acoustic comfort, PDIAQ = percentage of dissatisfaction in IAQ comfort, IEQ(PM) = IEQ performance, QT = thermal quality, QA = acoustic quality, QV = visual quality, QIAQ = IAQ.

4. Environmental data: T = temperature, globe T = globe temperature, radiant T = radiant temperature, RH = relative humidity, V-air = air movement, SPL = sound pressure level, I = illuminance level, CO2 = CO2 concentration, IBN = background noise index, IEN = effect of noise index, ISQ = sound quality index, IVP = voice perception index, ILQ = listening quality index, ALQ = artificial lighting quality index, ALS = artificial lighting sources index, ALG = artificial lighting annoying glares index, NLR = natural lighting reflections index, NLQ = natural lighting quality index.

5. Subjective data: S(T) = satisfaction with temperature, S(RH) = satisfaction with relative humidity, S(thermal) = satisfaction with thermal environment, S(SPL) = satisfaction with sound, S(sound privacy) = satisfaction with sound privacy, S(acoustic) = satisfaction with acoustic environment, S(I) = satisfaction with illuminance, S(visual) = satisfaction with visual environment, S(odor) = satisfaction with odor, S(IAQ) = satisfaction with IAQ.

6. The equation with “~” means the study only gave the coefficients of predictors, and no original model was provided.

5.2. Correlations between environmental factors and overall satisfaction through IEQ as mediators

Most of our reviewed studies fell into this category, which hierarchically quantified the interconnections between three levels. These studies first calculated occupants' comfort or evaluation about each IEQ domain, which was then aggregated to determine their overall satisfaction. We grouped the studies based on their modeling approaches.

5.2.1. Linear regression models

Mui and Chan [64] was a pioneering study which applied multivariate linear regression to describe occupants' overall satisfaction in offices through their thermal, acoustic, and IAQ assessment. Visual comfort was not included, as the illuminance variation was slight and within the acceptable range. This study calculated the percentage of dissatisfaction and divided it into three levels, which corresponded to offices above (10% dissatisfaction rate), equal to (20%), or below (30%) the average-grade office. Ncube and Riffat [74] developed a linear model for offices in the UK ($R^2 = 0.94$), which provided insights to reduce energy use without sacrificing overall satisfaction. This model indicated that IAQ had the highest impact on overall satisfaction: a 1-unit increment in IAQ comfort can enhance the overall satisfaction by 0.36, followed by thermal, acoustic, and visual comfort. Similarly, Cao et al. [55] conducted several field evaluations in offices, educational buildings, and libraries in China. The multivariate linear regression model showed that thermal comfort had the highest impact on overall satisfaction (coefficient: 0.316) while IAQ was the lowest (coefficient: 0.118). However, this model only applied to near-comfortable environments and may not be extrapolated to other conditions. Besides, Mihai and Iordache [58] used a linear model to represent the combined effect of thermal, acoustic, and visual comfort in educational buildings. The weight of each IEQ was directly obtained from occupants' self-reported data. This model categorized classrooms and offices into five levels based on the predicted overall satisfaction, which helped to identify areas that need renovations. Likewise, Buratti et al. [54] collected occupants' subjective responses and thermal, acoustic, and visual comfort. Higher weights were assigned to responses that showed strong correlations with experimental data. Nimlyat [23] studied the relationship between IEQ and overall satisfaction using the structural equation modeling (SEM). In their study, the latent variable IEQ performance was a mediator between overall satisfaction and four IEQ domains, which had a significant direct effect on satisfaction (loading: 0.83) and was affected by thermal (0.62), acoustic (0.50), visual (0.48), and IAQ (0.42).

5.2.2. Non-linear regression models

Contemporary studies also explored non-linear approaches, such as logistic regression and geometric mean model, to predict overall satisfaction. Wong et al. [75] adopted multivariate logistic regression to compare the importance of thermal, IAQ, acoustic, and visual comfort in air-conditioned offices in Hong Kong. In contrast to [75] which focused on offices, Lai et al. [76] conducted a similar experiment for residential buildings in Hong Kong. This study suggested that thermal and acoustic comfort were the most important contributors to overall satisfaction while IAQ was the least important one. A possible explanation of this result was the subtropical climate and high occupant density in Hong Kong. Tang et al. [25] suggested that linear models may isolate the contribution of each IEQ to overall satisfaction, regardless of their interactions. Subsequently, they proposed an arithmetic mean model (a type of linear model) and a geometric mean model (a non-linear model) which contained the interaction of three IEQ domains. Compared to the linear regression model, the geometric mean model showed higher prediction accuracy (reduced the residual standard error by 14%). This study indicated that overall satisfaction was more influenced by the worst IEQ domain. As such, strategies for improving the overall satisfaction should target it accordingly to improve the effectiveness of

intervention.

5.2.3. Other modeling approaches

Other modeling approaches have also been applied to interpret the relationships between IEQ and overall satisfaction, including but not limited to machine learning (ML), open IEQ acceptance model, Bayesian IEQ acceptance model, analytic hierarchy process (AHP), Kano's model of satisfaction, and ensemble averaging method. Specifically, Tang et al. [69] compared three types of models in terms of their accuracy and interpretability in overall satisfaction prediction, including linear (ordinary least squares regression), non-linear (weighted geometric mean model, weighted harmonic mean model), and ML (random forest and extreme gradient boost) models. Results demonstrated that the linear model was underperforming; the non-linear model performed better when only four IEQ were considered; while the ML models were more effective in more complicated settings in which large amounts of data and IEQ interactions were encountered. Artificial neural network (ANN), as an extension of linear or nonlinear regressions [78], has been applied in many studies. For example, a few studies [24,79] trained ANN models for occupants' thermal, acoustic, and visual comfort. By comparing model predictions with occupant feedback, they quantified the interaction and combined effects of IEQ domains on occupants' overall satisfaction. Besides, Wong et al. [80] proposed an open IEQ acceptance model using data from air-conditioned offices, residential buildings, and university classrooms. This approach had the flexibility of adding new IEQ domains or observations of environmental factors to update the model. The predicted overall satisfaction using this approach was consistent with their previous logistic regression model [75]; however, the results of individual IEQ acceptance were inconsistent. Building upon [80], Tsang et al. [71] proposed a Bayesian IEQ acceptance model to calculate overall satisfaction from binary IEQ responses in residential buildings. This approach aimed to update previous IEQ models using only a small amount of field data from a similar environment. Yang and Mak [61] employed fuzzy comprehensive evaluation (FCE) and analytic hierarchy process (AHP) to link the overall satisfaction, IEQ acceptance, and environmental factors. This study calculated the weights of IEQ (thermal > acoustic > visual > IAQ) and environmental factors and demonstrated the means to prevent overall satisfaction from being dominated by an extremely unfavorable IEQ (e.g., too hot or too noisy). Based on the feedback from 60 subject matter experts, Miao and Ding [81] used AHP to calculate the weights of four IEQ in six building types. They found thermal comfort and IAQ ranked consistently high in all building types except in the meeting rooms. Likewise, Li et al. [77] combined the subjective weighting from 405 experts using fuzzy hierarchy process (FAHP) and objective weighting from field measurements using entropy weighting method (EM). The resulting weighting scheme showed that thermal comfort ranked the highest, followed by acoustic comfort, IAQ, and visual comfort.

Researchers also made analogies between IEQ importance and consumers' preferences. Kim and De Dear [82] applied the Kano's model of satisfaction, which was initially proposed in marketing to prioritize product attributes according to their impacts on customer satisfaction [83], to evaluate the impacts of 15 environmental factors. The results showed that temperature, SPL, and building design (e.g., visual privacy, adjustability of furniture, colors, and textures) were the most impactful factors. IAQ and illuminance were found to be proportional factors, which indicated they had a linear relationship with overall satisfaction. In air terminals, Kano's model was used to determine the effect of IEQ on passengers' satisfaction [84]. In this context, thermal comfort and space layout were the most influential ones, while acoustic, visual, and IAQ comfort had linear proportional effects. Additionally, Wei et al. [85] applied ensemble averaging methods to compare the importance of IEQ across seven Green Building (GB) schemes (i.e., BREEAM, KLIMA, DGNB, ITACA, LiderA, LEED, NABERS) and 14 research publications. This study found that the weights of thermal and visual comfort were very close. However, discrepancies were found in acoustic comfort and

IAQ. Acoustic comfort was less important while IAQ was more important in GB schemes (thermal: 0.27, visual: 0.22, acoustic: 0.17, IAQ: 0.34) than they were in research publications (thermal: 0.28, visual: 0.21, acoustic: 0.26, IAQ: 0.25).

5.3. One-vote veto effect

Some studies observed a one-vote veto effect that occupants' overall satisfaction was significantly eroded if they were dissatisfied with an individual IEQ. This effect highlighted that some IEQ domains have predominant negative influences on overall satisfaction, regardless of the performance of other IEQ. For instance, Wu et al. [18] compared different IEQ combinations and concluded that discomfort with a single domain would lead to 65%–83% dissatisfaction with the overall environment. Cao et al. [55] found that extreme thermal and acoustic conditions can override satisfaction with other IEQ domains and thus seriously decrease overall satisfaction. However, this overriding effect was not captured by prediction functions. Similarly, one-vote veto effects were also reported in other studies [57,86]. Such findings indicated that the previously believed linear relationship between IEQ domains and overall satisfaction may be over-simplified. As such, non-linear and ML approaches that are capable of modeling more complex relationships in large data sets have considerable potentials to achieve better performance than linear models [69].

5.4. Summary of combined effects

In summary, the correlation between environmental factors, IEQ, and overall satisfaction has been quantified by linear, non-linear, and other modeling approaches. The inconsistent IEQ importance across studies (see Fig. 3) can be caused by the varying research designs (e.g., sample size, number of environmental factors, exposure time), building types, and modeling approaches (e.g., linear regression, geometric mean model). Thus, it is challenging to directly validate model performance, evaluate model adaptation to other datasets, and compare models across studies. Inspired by the global thermal comfort database [87], developing a similar global multi-domain IEQ database through international collaboration can help address those challenges. This effort would enable researchers and practitioners to share their data, develop IEQ models, benchmark model performance, and compare results for different contexts (e.g., locations, building types), and facilitate the development of building IEQ standards and guidelines and inform building-operations practices.

It is worth noting that studies reviewed above mainly focused on the combined effects of isolated IEQ domains without considering the

impacts of their interactions [20]. Such interactions have gained attention in recent years and were captured in a few most recent studies. For example, Tang et al. [25] included the interaction between SPL and illuminance in the acoustic model, and the interaction between thermal, visual, and acoustic comfort in the overall satisfaction model. Nimlyat [23] evaluated overall satisfaction through IEQ performance, which quantified the effects of thermal quality, acoustic quality, visual quality, IAQ, as well as the interactions among the environmental factors. Sun et al. [24] and Wu et al. [79] proposed ANN models for predicting the probability mass functions (PMFs) of sensation votes which incorporate the IEQ interactions into the overall satisfaction. Their conclusions are consistent with [25]. According to our review, those studies are mainly based on field evaluations conducted in near-comfortable environmental conditions. Compared to climate chambers, field studies are less controllable, which poses a challenge to quantify IEQ interactions. Thus, in the next section, we reviewed the findings of two and more IEQ interactions to highlight the potential effects that should be considered in overall satisfaction models.

6. The interaction effects between IEQ domains

In this section, we reviewed the studies that involved interactions of two or more IEQ domains and their impacts on occupants' IEQ comfort (see Table 3). Researchers have studied the effects of interaction on occupant performance [26,96,100–104], attention [93], and mood [105]. However, those findings are outside the scope of this review. Fig. 4 presents the number of publications for each type of interaction, either from climate chambers or field studies. In total, 55 relevant publications were reviewed, 23 of them were published after the previous review paper [20] and 10 of them were conducted in field studies. Publications which had more than one type of interaction were counted more than once. As shown in Fig. 4, interactions between thermal and visual (23), thermal and acoustic (19), thermal and IAQ (15), and acoustic and visual (13) were the most concerned areas while interactions between acoustic and IAQ (8), and visual and IAQ (5) were less explored. Six studies introduced three IEQ interactions. No study covered the interactions of all four IEQ domains.

6.1. Thermal and acoustic comfort

Effect of acoustics on thermal comfort. SPL, sound types, and the duration of sound stimuli can change occupants' sensitivity to temperature and humidity, and further alter their thermal sensation and comfort. Correlations have been found between a slightly higher SPL and the thermal comfort state, as some environmental controls, such as opening

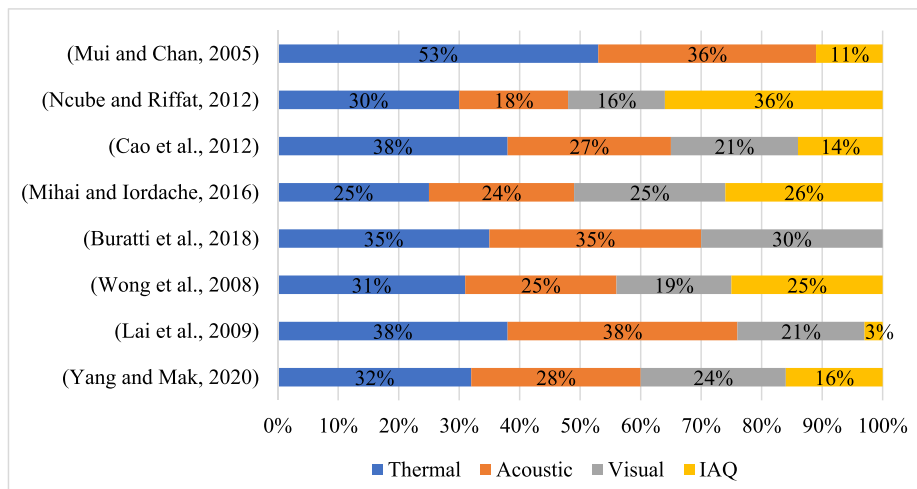


Fig. 3. IEQ weighting in some example publications.

Table 3
Summary of interaction effects

Study	IEQ domain	Experiment design	Occupant	Environmental data	Experiment duration	Statistical analysis	Modeling approach	Main findings
[40]	Thermal and acoustic; thermal and IAQ; acoustic and IAQ	Office (field)	69	T: -1.4 to 22 °C SPL: building1: <55 dB; building2: 60 to 65 dB	12 months	Inferential analysis	N/A	When environmental controls are available: 1) Noisy condition is associated with cold sensation, vice versa. 2) Cool condition is associated with good IAQ, vice versa. 3) Quiet condition is associated with good IAQ, vice versa.
[57]	Thermal and acoustic; thermal and visual; acoustic and visual	Office (field)	120 students (60 M, 60 F) Age: 22 Ci: 0.45 clo	T: 18 to 30 °C V-air: 0.02 to 0.42 m/s RH: 40% to 65% SPL: 45 to 65 dB I: 100 to 1,300 lx	145 min	Chart analysis	N/A	1) When occupants are thermally satisfied, they have a higher acceptable SPL range. When the SPL decreases, they have a wider acceptable temperature range. 2) Occupants prefer a higher illuminance and a lower SPL when they are thermally satisfied. 3) Occupants prefer a higher illuminance and a moderate temperature when they are acoustically satisfied. 4) Thermal and acoustic comfort have one-vote veto power over the overall satisfaction.
[59]	Thermal and acoustic	Climate chamber	108 students (54 M, 54 F) Age: 23.5 (M); 22.9 (F) Ci: 0.6 clo	T: 14 to 34 °C V-air: 0.1 m/s SPL: 35 to 85 dB	120 min	ANOVA; posteriori Newman-Keuls test; chi-square test	N/A	1) Occupants in a thermal-neutral condition would evaluate SPL more severely. 2) Occupants in cool or warm conditions would be less affected by SPL due to the priority of thermal perception. 3) To alleviate thermal discomfort, occupants would switch on ventilation or open the window to control temperature, which may increase SPL and decrease acoustic comfort.
[88]	Thermal and acoustic	Climate chamber	26 occupants (13 M, 13 F) Age: 25.5 (M); 25.8 (F) Ci: 0.73 clo	T: 18, 24, 30 °C RH: 30%, 60% for 18 °C; 27%, 43%, 65% for 24 °C; 30%, 60% for 30 °C V-air: 0.1 m/s Ventilation rate: 110 m ³ /h I: 880 lx Sound type: fan noise; babble noise SPL: 45 to 65 dB	90 min	ANOVA	N/A	1) Thermal sensations are not affected by 3-min exposure to indoor SPL (45 to 65 dB). 2) Only babble noises are perceived as louder and more annoying at high temperature with low RH. Such perception declines with the increasing temperature and RH. However, thermal condition does not affect fan noises perception.
[30]	Thermal and acoustic; thermal and visual; acoustic and visual	Classroom (climate chamber)	60 students (30 M, 30 F) Age: 23.3 (M); 21.3 (F) Ci: 0.75 clo	T: 20, 25, 30 °C RH: 45% V-air: < 0.1 m/s I: 150 lx; 500 lx; 1,000 lx Sound type: babble; fan; music; water SPL: 45 to 75 dB	145 min	ANOVA; Bonferroni's post-hoc test; partial eta squared	N/A	1) Thermal comfort increases when the SPL decreases at 500 lx illuminance level. However, the thermal-visual interaction based on illuminance is unclear. 2) Visual comfort increases at lower SPL and 25 °C. However, the effect of acoustics on visual comfort based on illuminance is unclear. 3) Acoustic comfort increases in thermal-neutral conditions (25 °C). 4) When SPL increases from 35 to 75 dB, the effect of SPL on thermal comfort is greater than the effect of illuminance (150 to 1,000 lx) on thermal comfort. 5) Overall satisfaction increases when SPL decreases in thermal-neutral and bright conditions.

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Table 3 (continued)

Study	IEQ domain	Experiment design	Occupant	Environmental data	Experiment duration	Statistical analysis	Modeling approach	Main findings
[56]	Thermal and acoustic	Office (climate chamber)	16 students (8 M, 8 F) Age: 18.9 (M); 21.3 (F) Ci: 1.2 clo	T: 18, 22, 26 °C RH: 50% V-air: < 0.1 m/s Sound type: music; fan SPL: 50 dB; 65 dB; 80 dB	155 min	ANOVA; <i>t</i> -test	N/A	1) Thermal comfort is affected by temperature and SPL, but not affected by temperature × SPL. 2) Compared to fan noise, music can improve thermal comfort, and this difference is significant between fan noise and music under 65 dB. 3) Acoustic comfort is only significantly affected by SPL and sound type. It is not affected by temperature or temperature × SPL. 4) The effect level of environmental factors on overall satisfaction is SPL > temperature > temperature × SPL.
[17]	Thermal and acoustic; thermal and visual; thermal and IAQ	Office (field)	21 occupants (12 M, 9 F) Age: 21.7±2.6 Ci: 1.15 clo	T: 16 to 28 °C Controlled variables: globe T; RH; I; SPL; CO2	120 min, last for one week	ANOVA; <i>t</i> -test; Spearman or Pearson correlation test	N/A	1) When thermal environment is unsatisfying, it weakens the comfort expectations of other IEQ domains, resulting in less dissatisfaction with other IEQ, and vice versa.
[90]	Thermal and visual	Office (climate chamber)	75 students (45 M, 30 F) Age: 18 to 30 Ci: 0.7 clo	T: 19, 22, 26 °C I: 136 to 1,043 lx Daylight color: blue; orange; neutral	180 min/session	ANOVA; post-hoc test; variance inflation factors (VIF)	Linear mixed model	1) At 22 °C and 26 °C, blue daylight induces a colder sensation and decreases thermal comfort. While orange daylight induces a warmer sensation and improves thermal comfort. 2) At 19 °C and 22 °C, occupants prefer a higher temperature under blue daylight compared to the orange one. 3) Both colored daylight and temperature affect overall satisfaction, but the interaction (colored daylight × temperature) is insignificant. 30-min exposure influences occupants' responses: they are less affected by colored daylight after 30-min exposure compared to the beginning; the impact of temperature increases through the exposure.
[89]	Thermal and visual	Office (field)	Summer: 108 occupants (57 M, 51 F) Winter: 160 occupants (83 M, 77 F)	Summer: T: 24.2 ± 0.9 °C I: 706 ± 659 lx (desk level); 540 ± 475 lx (eye level) Winter: T: 24.4 ± 1.3 °C I: 883 ± 768 lx (desk level); 641 ± 488 lx (eye level)	N/A	Kruskal-Wallis test; post-hoc test; Dunn test; ANOVA	N/A	1) Thermal sensation is only affected by temperature, not by illuminance, both in summer and winter. 2) Thermal comfort is affected by illuminance in both seasons, especially at a higher temperature (> 25 °C), thermal comfort decreases at low illuminance level (< 300 lx). 3) Illuminance level alters thermal expectation and may affect thermal comfort, occupant may prefer a cooler condition in dim light (< 300 lx).
[91]	Thermal and visual; visual and IAQ	Office (climate chamber)	44 occupants (28 M, 16 F) Age: 23.1 (M); 24.9 (F) Ci: 0.5 clo, 1.0 clo	T: 19 °C, 22 °C, 27 °C Color temperature: 2,700 to 6,300 K I: 1,000 lx (at desk level) V-air: 0.1 m/s SPL: 48 dB	120 min	N/A	Generalized estimating equations (GEE); linear mixed model	1) At 22 °C (thermal-neutral), increasing color temperature would decrease thermal sensation. 2) In ideal conditions (without the influence of other visual factors), increasing color temperature from 2,700 K to 6,200 K at a comfortable temperature is equivalent to a temperature increase of 1.7 °C. 3) At 22 °C, increasing color temperature would improve IAQ comfort. 4) Color temperature has no effect on the performance of concentration test.
[92]	Thermal and visual	Climate chamber	42 occupants (24 M, 18 F) Age: 23 Ci: 0.54 clo, 0.57 clo	T: 22 °C Color temperature: 1,772 K; 4,000 K; 11,530 K I: 500 lx	120 min	Friedman test; Wilcoxon signed-rank test; partial F test	N/A	1) In summer, applying cold, neutral, and warm light while increasing temperature by 1.25, 0.46, and 0.23 °C respectively can maintain occupant thermal comfort and save energy.

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Table 3 (continued)

Study	IEQ domain	Experiment design	Occupant	Environmental data	Experiment duration	Statistical analysis	Modeling approach	Main findings
[93]	Thermal and visual	Climate chamber	33 students (17 M, 16 F) Age: 22.1 Ci: 0.8 clo	T: 22, 37 °C WBGT: 18.6 to 30 °C RH: 15% to 25% V-air: 0.1 m/s I: 200 lx; 500 lx; 1,500 lx Color temperature: 4,500 K	90 min	ANOVA; <i>t</i> -test	N/A	1) At 22 °C, the highest attention is from the exposure to 1,500 lx. Attention has a positive correlation with the illuminance level. 2) At 37 °C, the highest attention is from the exposure to 200 lx. Attention has a negative correlation with the illuminance level. 3) At 22 °C, increasing illuminance level would increase the average number of correct responses, and decrease the average response time, commission error, and omission error. 4) At 37 °C, increasing illuminance level would decrease the average number of correct responses, and increase the average response time, commission error, and omission error.
[94]	Thermal and IAQ	Climate chamber	124 students (62 M, 62 F)	T: 20 to 28 °C RH: 30%; 40%; 70% V-air: < 0.1 m/s; 0.2 to 1 m/s Air quality: low pollution, high pollution	60, 165, 180, 235 min	ANOVA and Newman Keuls test (normally distributed data); Friedman ANOVA and Wilcoxon matched-pair test (other data)	N/A	1) Increasing air movement can reduce the negative effect of the increased temperature (from 20 to 26 °C), RH (from 30% to 60%), and pollution on air acceptability and perception of air freshness. However, it did not affect odor intensity. 2) Improvement in IAQ comfort due to air velocity is more pronounced in warm and humid conditions and at high background pollution levels.
[25]	Thermal and IAQ; acoustic and visual; acoustic and IAQ	Office (climate chamber)	8 students (4 M, 4 F) Age: 23 Ci: 0.5 clo	PMV: -1.5 to 1.5 V-air: < 0.2 m/s SPL: 40 to 65 dB I: 100 to 1,000 lx Color temperature: 4,000 K CO ₂ : 500 to 1,500 ppm	180 min/session	Partial correlation analysis	Multiple linear regression; multiple nonlinear regression	1) Darker environments lead to quieter perceptions and increase acoustic satisfaction. 2) IAQ comfort decreases in warmer conditions (PMV > 0). 3) A slightly negative effect of SPL on IAQ comfort is observed.
[95]	Thermal and IAQ	Office (climate chamber)	12 students (6 M, 6 F) Ci: 0.9 clo	T: 22, 30 °C SPL: 50 dB Ventilation rate: 10 l/ (s × per)	270 min	ANOVA; <i>t</i> -test	N/A	1) Compared to 22 °C, occupants under 30 °C perceive the IAQ to be worse; The intensity of SBS, fatigue, mental workload, and negative mood disturbance all increase. Occupants are less willing to exert effort in working, and the performance of simulated office work and neuro-behavioral tests decreases. 2) At 30 °C, heart rate, respiratory ventilation rate, and ETCO ₂ concentration all increase; SPO ₂ concentration decreases; tear film quality reduces, and eye discomfort is intensified.
[22]	Thermal and IAQ	Office (climate chamber)	18 occupants (9 M, 9 F) Age: 20 to 26	T: 20 to 30 °C RH: 40% to 90% Ventilation rate: 6 to 25 l/ (s × per) SPL: 52.2 ± 4.1 dB I: 468 ± 46 lx for daytime; 284 ± 21 lx for night	70 min/session, last for 2 weeks	Response surface methodology (RSM)	Multiple quadratic regression orthogonal model	1) The interaction of RH and ventilation only occurred at a higher temperature. In high temperature and high RH conditions, sufficient fresh air can change subjective perception; while in high temperature and low RH conditions, increasing fresh air can improve the performance of neuro-behavioral tasks. 2) The impact of RH changes on work performance under higher temperature is about 4% to 10% greater than that under low temperature. No consistent change in performance is observed when temperature is below 25 °C. 3) Performance loss due to a 1 to 2 °C increase in temperature can be compensated by increasing ventilation rate to 25 l/ (s × per).

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Table 3 (continued)

Study	IEQ domain	Experiment design	Occupant	Environmental data	Experiment duration	Statistical analysis	Modeling approach	Main findings
[96]	Thermal and acoustic; thermal and IAQ	Office (climate chamber)	56 students (30 M, 26 F) Age: 24	T: 23.5 °C; 28 °C SPL: 52 dB; 60 dB Sound type: traffic noise Air quality: pollution load absent; pollution load present	20 min/ session, last for 3 h	T-test; Wilcoxon matched-pair signed-rank test	N/A	1) Warm noisy conditions, warm polluted conditions, warm noisy and polluted conditions all decrease the performance of addition test.
[24]	Thermal and acoustic; thermal and visual; acoustic and visual	Office (climate chamber)	35 students (8 M, 27 F) Ci: 0.75 clo	T: 18 to 28 °C RH: 45% V-air: < 0.1 m/s SPL: 35 to 60 dB I: 180 to 1,380 lx Color temperature: 6,500 K	60 min, last for 8 days	N/A	ANN	1) For thermal: increasing SPL contributes to colder feelings when temperature is lower than neutral (23 to 24.7 °C), this effect is significant; it causes cooler feelings when temperature is higher than neutral. Higher illuminance leads to colder or warmer feelings depending on the temperature. 2) For visual: increasing temperature or decreasing SPL leads to dimmer feelings (illuminance < neutral level of 580 to 650 lx); decreasing temperature or SPL leads to a brighter feeling (illuminance > neutral level). These effects are slight. 3) For acoustic: decreasing temperature or increasing illuminance leads to slightly noisier sensations (SPL < neutral SPL of 41.5 to 43.5 dB). These effects are slight.
[97]	Thermal and visual; thermal and acoustic; thermal and IAQ; visual and IAQ; acoustic and IAQ	Classroom (field)	55 groups (No. of occupants are not mentioned)	T: 20.4 to 24.7 °C SPL: 61 to 75 dB I: 20 to 1,000 lx CO2: 200 to 1,400 ppm	60 min	Permutation test	Regression analysis	1) Higher temperature reduces the perceived IAQ; higher CO2 level makes occupants feel warmer. 2) Increased temperature is associated with a small reduction in visual comfort; illuminance is significantly correlated with thermal comfort. 3) Increased temperature is associated with a small reduction in acoustic comfort; higher SPL is associated with cooler sensation. 4) Higher CO2 level does not affect visual or acoustic comfort. However, illuminance has a significant effect on IAQ.
[53]	Thermal and visual; acoustic and visual	Shopping center (climate chamber)	27 students (13 M, 14 F) Age: 21.2 ± 3.26 (M); 20.2 ± 3.38 (F)	T: 23 °C SPL: 75 dB I: 200 to 1,500 lx Color temperature: 3,000 K; 4,500 K; 6,000 K	20 min/ session, last for 3 weeks	Pearson correlation analysis	N/A	1) Pearson correlations between visual and thermal comfort, and visual and acoustic comfort are significant. 2) When visual comfort changes from “dissatisfied” to “satisfied”, the thermal and acoustic dissatisfaction rates reduce by 34.47% and 17.74%.
[98]	Acoustic and visual	Residential building (climate chamber)	30 occupants (12 M, 18 F) Age: 23.8	SPL: 35 to 65 dB Color type: red; yellow; cyan; green Brightness level: high; medium; low Indoor scene: study room; living room	20 s/acoustic-visual stimulus	ANOVA	N/A	1) The effect of brightness on acoustic comfort under traffic noise is more significant than the effect of color: low brightness increases subjective noise annoyance; color has no influence on noise annoyance in a medium brightness condition; interaction of color and SPL affects noise annoyance regardless of the brightness condition. 2) A reasonable combination of color and brightness can improve visual and acoustic comfort.
[99]	Acoustic and IAQ	Climate chamber	9 occupants Age: 25.6	T: 21.6 ± 0.5 °C RH: 45% ± 1.1% Ventilation rate: 10 to 15 cm/s SPL: 45 dB; 75 dB Odor: coffee aroma	135 min	MANOVA	Repeated-type general linear model	1) For acoustic comfort, exposure to odor causes neither synergism nor antagonism effect: odor does not affect acoustic comfort. 2) For IAQ, there is an insignificant antagonistic effect of SPL on the perceived odor: increasing SPL decreases the odor-related discomfort.

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Table 3 (continued)

Study	IEQ domain	Experiment design	Occupant	Environmental data	Experiment duration	Statistical analysis	Modeling approach	Main findings
[27]	Three IEQ: acoustic, visual, and IAQ	Classroom (climate chamber)	250 students	T: 21 °C SPL: 0 to 100 dB I: soft; direct; indirect light Ventilation: mixing; displacement; natural ventilation	35 min/session, 3 sessions	ANOVA; <i>t</i> -test; chi-square test	Stepwise multi-regression	1) Occupants perceive the coldest temperature in quiet environment, and neutral when background conversation exists. 2) Occupants perceive the worst odor when background conversation exists and vice versa. 3) Occupants perceive the worst odor in direct light and the best in indirect light. 4) The interaction of lighting and SPL on odor perception is significant.
[28]	Three IEQ: thermal, acoustic, and IAQ	Office (field)	154 occupants	Focus group (12 occupants) T: 24.2 °C RH: 33% SPL: 47 dB	N/A	Goodman Kruskal's gamma; Kendall's correlation	Mixed effect logistic regression	1) Thermal comfort is associated with both IAQ and SPL, but their interactions on thermal comfort is not evident. Thermally uncomfortable occupants perceive worse IAQ. 2) Acoustic comfort is associated with operative temperature. 3) Perceived IAQ is associated with operative temperature and SPL.
[6]	Three IEQ: thermal, acoustic, and visual	Classroom (climate chamber)	158 students (95 M, 62 F) Age: 17 to 49	Normal zone: T: 22 °C I: 500 lx SPL: 35 dB Outside of comfort zone: T: 26 °C I: 2,500 lx SPL: 65 dB (reading), 60 dB (listening)	N/A	ANOVA	N/A	1) Compared to normal zone, occupants doing listening tasks outside of comfort zone have lower scores. However, this difference is not detected for occupants doing reading tasks. 2) Occupants outside of comfort zone report that temperature and sound have larger negative effects on performance and mood than those in normal zone. 3) Compared to occupants doing listening tasks, those doing reading tasks are more likely to attribute their poor performance to SPL.

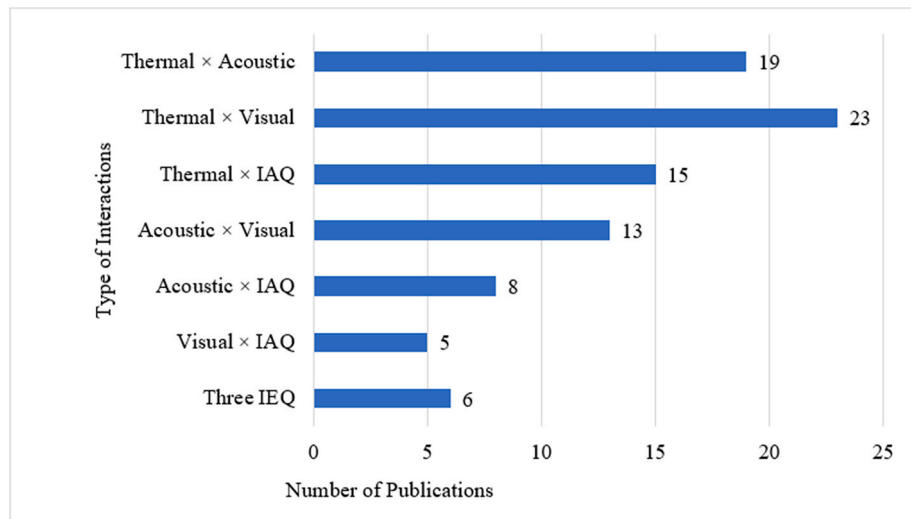


Fig. 4. Number of reviewed publications for each interaction effect.

windows and turning on air conditioner, may increase SPL [40,59,97]. A quiet environment (i.e., low SPL) would sharpen occupants' sense of temperature (also known as the "masking effect" [27,106]), widen their acceptable temperature range and make them more likely to feel thermally comfortable [57]. In terms of sound types, Guan et al. [56] found that 65 dB pure music was an effective mood-induction source to induce a wider comfortable temperature range compared to fan noise of the same SPL. Geng et al. [107] supplemented that such interaction was only significant in the Spring based on their investigation of open-plan spaces on campus. In this study, broadcasting music can improve thermal comfort in all thermal environments (slight cold stress, no thermal stress, and moderate heat stress). Running water and bird sounds can also improve thermal comfort compared to crowd sound in cold environments. For the duration of sound stimuli, Yang et al. [60] suggested that 3 min of exposure to moderate noise (45–65 dB) would not affect thermal sensation. Similar findings were also reported in Fanger et al. [108] (2.5 h of exposure to 40/85 dB white noise), Nagano and Horikoshi [109] (2 min of exposure to 46.8 dB air conditioning sound or 59.2/73.1/80.0/95.4 dB traffic sound), Witterseh et al. [110] (3 h of exposure to 35 dB quiet background or 55 dB open-plan office sound).

Effect of thermal on acoustic comfort. Findings in this area were inconsistent. Pellerin and Candas [59,111] observed that occupants in thermal-neutral conditions were more affected by SPL: they perceived higher SPL at 24 °C than at 18 °C and 30 °C. However, several studies argued that a moderate temperature would contribute to acoustic comfort [57], and increased temperature would make occupants more sensitive to various sounds (e.g., crowd noise, machine noise, wind and insects) and lower their acoustic comfort despite the same SPL [97,107]. Nevertheless, some studies [56,112] suggested that acoustic comfort was not significantly affected by temperature or the interaction of temperature and SPL in their investigated temperature range (18 °C–26 °C). Some studies argued that temperature and humidity affected the perception of sound types to different extents. Yang et al. [88] observed that occupants perceived louder babbling sound and were more annoyed under increased temperature (18 °C–30 °C) and 30% RH, whereas such effect decreased under 60% RH. However, no interaction between temperature and humidity was observed under fan noise.

6.2. Thermal and visual comfort

Effect of visual on thermal comfort. This area has been intensively studied. The frequently discussed visual factors were illuminance level and light color. Some studies [24,53] have reported that visually satisfied occupants perceived better thermal comfort in the same

environment, while a few studies did not observe any significant effects [30,113,114].

Illuminance affects occupants' thermal sensation, satisfaction, sensitivity, and thermo-physiological responses (e.g., thermo-regulation, melatonin secretion). Chinazzo et al. [89] discovered that occupants in dim light (<300 lx) were less thermally satisfied at high temperature (above 25 °C), which implied occupants' thermal expectations induced by different illuminance levels may affect their thermal comfort. Yamazaki et al. [106] simulated a typical office environment and reported occupants' temperature sensitivity was low in poor lighting and would increase with higher illuminance. In a review study, Te Kulve et al. [115] showed that bright light (e.g., 1500 lx, 2500 lx, 3000 lx) or light with shorter wavelengths (6500 K) reduced melatonin secretion in the evening, which delayed the natural decline in core body temperature as compared to dim light (e.g., <50 lx) or light with longer wavelengths (3000 K). While in the morning, bright light facilitated the decline in melatonin levels and thus accelerated the increase of core body temperature. As a result, occupants felt warmer in bright light as compared to dim light. The authors' follow-up experimental study [116] confirmed that higher visual comfort can partially compensate thermal discomfort during a 225-min light exposure. It is worth mentioning that most selected studies investigated illuminance-thermal interactions in short-term exposures (e.g., 10–15 min). It is unclear whether the observed effects still exist in long-time exposures. Meanwhile, the perceived comfortable illuminance level varies greatly across occupants, so personal preferences should be considered in this type of interaction.

Light colors can also induce different thermal sensations and preferences [117]. After studying occupants' responses at 22 °C and 26 °C, Chinazzo et al. [90] found that occupants felt colder in blue daylight and warmer in orange daylight as compared to neutral daylight. It has also been documented that occupants preferred higher temperature under blue light compared to orange light [90] or red light [108,115,118]. These findings indicated a non-negligible psychological effect of light color. Moreover, increasing the color temperature from 2700 K (warm white) to 6200 K (cold white) would decrease thermal comfort [91,119], which was equivalent to the impact of a temperature change of 1.7 °C [91]. Likewise, Golasi et al. [92] showed that applying cold (11,530 K), neutral (4000 K), and warm (1772 K) light while increasing temperature by 1.25 °C, 0.46 °C, and 0.23 °C, respectively can maintain the same thermal comfort level. As such, Toftum et al. [91] suggested that adjusting light color from 4500 K to 2700 K can result in 8% savings in building energy use while still maintaining thermal comfort.

Effect of thermal on visual comfort. An early study [120] didn't observe any effect of thermal conditions on visual comfort. However, recent

studies showed that visual comfort can be affected by thermal conditions from three aspects: illuminance sensation, light color sensation, and glare sensation. Higher air temperature was observed to be associated with lower visual sensation votes [97] and a dimmer feeling if the illuminance was below the neutral point (580–650 lx), and vice versa [24]. When occupants were thermally comfortable, they showed better acceptance of high illuminance [57]. Temperature influenced occupants' light color evaluation as well. In lower temperature (19 °C), daylight tinted by the blue glazing was evaluated less comfortable than by the orange glazing [121]. Garreton et al. [122] observed a significant relationship between temperature and glare sensation ($r > 0.68$, $p = 0.001$). This study showed that occupants assessed their glare sensation to be uncomfortable once they were out of the thermal comfort zone.

6.3. Thermal and IAQ comfort

Effect of IAQ on thermal comfort. IAQ was found to be highly correlated with thermal comfort [25,28]. In particular, polluted air through a used carpet [94] or higher CO₂ concentration [97,123] can make occupants feel slightly warmer or more sensitive to heat while the temperature is unchanged. Bourikas et al. [28] explained this effect as “stuffiness”, a condition that is often resulted from low air movement, high CO₂ concentration, and high RH.

Effect of thermal on IAQ comfort. The indicators selected in this area mainly include temperature, RH, and air movement. Some studies [17, 124] found that occupants tended to assess IAQ more stringently under uncomfortable thermal conditions. Field studies in office buildings and classrooms [28,40,97] showed that higher temperatures significantly lowered occupants' IAQ comfort and acceptability, which was consistent with several prior studies [19,25,96,125–128]. Fang et al. [126, 127] supplemented that the significance of such interaction diminished with increased air pollution. Melikov and Kaczmarczyk [94] proposed that the negative influence of high temperature, humidity, and pollution level would be mitigated by increased air movement. In addition to higher temperatures, Lan et al. [19] suggested other possible contributors to lowered IAQ comfort including the increased CO₂ concentration, heart rate, and respiratory rate.

6.4. Acoustic and visual comfort

Effect of visual on acoustic comfort. Visual-acoustic interactions have been investigated with regard to realistic environmental factors, such as casual conversation, broadband low-frequency noise (e.g., background noise between 10 and 250 Hz), and artificial lighting. An early study [129] discovered an asymmetric interaction, i.e., visual factors had modest or no effect on acoustic pitch and loudness discrimination while acoustic factors had little or very little effect on visual brightness discrimination. Dang et al. [53] showed that changing visual evaluation from “dissatisfied” to “satisfied” alone can reduce acoustic discomfort by 17.74% in a simulated shopping center experiment. Acoustic sensations also varied with different illuminance levels. Occupants perceived a quieter surrounding in dark environments and slightly noisier when illuminance increased [24,25]. Meanwhile, color and illuminance also affected noise annoyance. Ma and Nie [98] examined 96 visual-acoustic stimuli and suggested that a high illuminance level can reduce subjective noise annoyance under warm color light (e.g., red, yellow), while a medium illuminance level had the same effect under cold color (e.g., green, cyan). Despite the above results, several other studies [130,131] did not observe a strong sensory enhancement in acoustic comfort by altering visual factors.

Effect of acoustics on visual comfort. There was limited literature in this area and the findings were fragmented. Although some researchers suggested that sound exposure did not increase occupants' perceived brightness [129,132] and visual comfort was predominantly determined by illuminance level [25], others indicated that acoustics can affect the illuminance acceptability, brightness perception, and relaxation of eyes.

For example, Huang et al. [57] observed that occupants were more satisfied with a higher level of illuminance when they were acoustically comfortable. Sun et al. [24] found that a lower SPL contributed to a dimmer feeling when illuminance was lower than the neutral point (580 lx–650 lx), and a brighter feeling was induced if illuminance was higher than the neutral point. Yang and Moon [131] observed the effect of SPL on the relaxation of eyes, but no relationship between SPL and brightness perception was reported. Shams et al. [133] found that a variable number of beeps can alter the visual perception and induce illusory flashing. However, the effect of acoustics on other visual factors, such as light color, has not been found in the reviewed studies.

6.5. Acoustic and IAQ comfort

Effect of IAQ on acoustic comfort. This area is rarely reported. Current studies attempted to connect odor and olfactory comfort with noise perception and acoustic comfort. Pan et al. [99] observed that odor had neither synergistic nor antagonistic effect on SPL perception, suggesting that it played little effect on acoustic comfort. CO₂ concentration changes also didn't show any meaningful effect on acoustic comfort [97]. Lee and Aletta [134] advocated that design protocols to control odor in open-plan offices (e.g., deck-to-deck partitions, separate exhaust, interstitial rooms, vestibules, hallways) may be beneficial to isolate sound and improve acoustic comfort.

Effect of acoustics on IAQ comfort. Pan et al. [99] indicated that noise would reduce odor intensity and IAQ discomfort, and ease occupants' needs for higher ventilation rates. Nevertheless, studies [25,27,28,40] observed that SPL negatively affected IAQ comfort, i.e., quiet environments were associated with good perceived IAQ, and vice versa. It was also reported by both field [3] and lab experiments [27] that occupants are more sensitive to sound and odor under the influence of background conversation.

6.6. Visual and IAQ comfort

In our reviewed studies, the IAQ acceptability was positively affected by the illuminance level [97,135], color temperature [91], and light color [136]. Pittana et al. [97] suggested that the effect between illuminance and IAQ is asymmetric. In particular, the perceived odor is the worst in direct light, and otherwise in indirect light [27]. Toftum et al. [91] noted that the perceived IAQ was fresher at 22 °C with an increasing color temperature of white LED lighting from 2700 K (warm white) to 6200 K (cool white). A relevant study about the aircraft cabin reported that passengers perceived better air quality in blue light than in yellow light [136].

6.7. Interaction effects among three and four IEQ domains

Existing studies primarily focused on two IEQ interactions while a holistic evaluation of three and more IEQ domains is lacking. To the best of our knowledge, there are six studies about three IEQ interactions. The interaction of temperature and illuminance on acoustic comfort was found statistically significant [18]. Meanwhile, Yang and Moon [30] observed occupants' overall satisfaction increased when SPL decreased in thermal-neutral (25 °C) and bright conditions. Decreased SPL can improve thermal comfort at 500 lx and lowered SPL can increase visual comfort at 25 °C. Bluyssen et al. [27] found the interaction of illuminance and SPL significantly affects the assessment of odor. Bourikas et al. [28] assessed the interactions of thermal, acoustic, and IAQ on occupants' satisfaction in three mix-mode office buildings. However, IAQ and SPL interaction had no significant effect on thermal comfort. Moreover, Marchand et al. [6] investigated students' reading and listening performance in a normal zone (22 °C, 500 lx, 35 dB) and a just outside comfort zone (26 °C, 2500 lx, 65 dB for reading, 60 dB for listening). This study showed that relatively higher temperature, illuminance, and SPL reduced listening performance while they had no

impact on reading. This result can be explained by the different effects of environments on occupants' distraction and cognitive load, i.e., occupants can filter out irrelevant negative environmental stimuli in a moderately difficult reading task, but would be affected in listening tasks due to declined attention [137]. Likewise, significant impacts of thermal, acoustic, and visual interaction on occupants' learning efficiency were also found in a recent study [29].

There are no conclusive interactions between the four IEQ domains. This may be attributed to the complex experiment design: 1) both IEQ and environmental factors are multi-faceted. Therefore, selecting the representative factors and their experimental ranges to reach statistically significant results is not an easy task. Since most studies adopted the factorial design, the experiment scenarios would increase exponentially as more factors are involved. Furthermore, it is challenging to recruit a large number of participants of different ages, genders, and cultural backgrounds [20]; and 2) participants may be unconsciously influenced by previous sessions in short-term exposure experiments. It is also possible that they are less affected by some environmental factors (e.g., light color) and more affected by others (e.g., temperature) in long-term exposures, thus impairing the validity of the results. If a longer exposure time is selected, the total experiment time will increase accordingly, leading to experiment fatigue or learning effect. Moreover, multi-collinearity is another challenge which can cause wrong estimation of the standard errors of model coefficients and misinterpret the weights of IEQ and their interactions [138]. Researchers are suggested to use the variance inflation factor (VIF) to detect multi-collinearity across multiple IEQ domains [90]. Apart from the challenges in climate chamber tests discussed above, field evaluations often have more uncontrolled variables that can concurrently influence the results, such as uncontrolled daylight, surrounding noise, and odor.

6.8. Summary of interaction effects

Three observations can be drawn from the selected studies: 1) some interactions are valid when the environmental factors are within the pre-determined experimental ranges. The results cannot be extrapolated beyond those ranges. For example, Geng et al. [84] suggested that unsatisfying thermal conditions in offices can weaken an occupant's expectations of acoustic, visual, and IAQ comfort, leading to less dissatisfaction with those IEQ and vice versa. This conclusion was valid when the temperature varied between 16 and 28 °C. Toftum et al. [91] showed that color temperature was associated with thermal sensation and IAQ only in thermal neutral environments (22 °C); 2) the exposure time is another influential factor, especially in climate chambers where occupants are exposed to different test scenarios in a short period. Yang and Moon [30] showed that occupants in short exposures (25 s of sound stimuli) were likely to compare their current environment with the most recent ones. Thus, their perceptions can differ from those obtained from a sustained exposure [25]. Chinazzo et al. [90] suggested that occupants' adaptation to experimental conditions can alter their responses. In their study, occupants were less affected by the colored daylight but more so by temperature at the end of a 30-min daylight exposure compared to the beginning; and 3) the significance level varies across different types of interactions. Interactions between temperature and lighting color were relatively insignificant compared to those between temperature and SPL [30,40].

7. Discussion

Over the past two decades, although a growing body of literature has investigated the relationships between environmental factors, IEQ comfort, and overall satisfaction, the findings of both interaction and combined effects are inconsistent and somewhat contradictory, highlighting the urgent need for systematic guidelines and recommendations on multi-domain IEQ research design and reporting [15]. Besides, very few studies (6) have investigated three IEQ interactions while no studies

addressed four IEQ interactions. In the following subsections, we discussed the possible causes of such results.

7.1. Generalized models versus personal models

Existing overall satisfaction and IEQ models are mainly generalized models which are developed based on the average responses of a group of occupants. Thus, such models are generic and unable to capture individual differences [139]. Some recent models, such as the open IEQ acceptance model [80] and Bayesian IEQ acceptance model [71], are able to perform model updates using new environmental data collected from similar environments. Nonetheless, personal data (e.g., gender, culture, age, health status) that are relevant to individual comfort still cannot be readily incorporated [139]. Therefore, generalized models may not accurately reflect an individual's needs and preferences [73, 139]. Besides, generalized models also have limitations in 1) model adaptation: those models are developed using fixed sets of data collected in a particular setting. Thus, they cannot directly adapt to different settings (e.g., different building types, populations) even with the new training data [69]; and 2) demanding development efforts: training data in generalized models are often measured by special sensors or instruments (e.g., air flow rate, mean radiant temperature), which are costly and difficult to obtain in real-world buildings.

On the other hand, personal models can effectively capture the potential variances among occupants [140,141]. Personal models are developed using individual feedback under different occupant and environmental conditions. Those models can be trained using various approaches and easily updated by new input data. Personal models have shown higher accuracy than generalized models [140,142]. However, their applications have been limited to isolated IEQ domains, such as thermal comfort [140,142–145], visual comfort [146,147], IAQ [148], and biophilia [38,149]. An expanded personal model that encompasses IEQ interactions and their impacts on overall satisfaction has not been realized yet. One challenge of personal models is the large amount of training data required from each occupant [140]. Insufficient data often lead to overfitting or the "cold start" problem. To increase the amount of data from each occupant, non-intrusive methods have been proposed, including using thermal cameras [142], web cameras [150], and wearable sensors [73,151,152]. Recent studies have also investigated means to cluster pre-trained personal models based on occupant similarities and ensemble those clusters for predicting a new occupant's thermal comfort [153]. This approach has demonstrated higher accuracy than conventional models and has potentially addressed the data inadequacy challenge.

Current building standards typically define the acceptable ranges of environmental parameters using results of generalized models with the aim to satisfy the majority of occupants. However, numerous field evaluations have suggested that current building operation practices are far from satisfactory (e.g. [154]). Thus, incorporating personal models, especially personal overall satisfaction models (which are currently lacking), into building standards will provide opportunities to enable human-centered building operations and accommodate context- and occupant-specific comfort requirements [145,155], thereby making buildings more responsive and adaptive to occupants' needs and expectations.

7.2. Climate chamber experiments versus field evaluations

Most studies on interaction effects are conducted in climate chambers in which the cubicles or rooms are well insulated and highly instrumented. The environmental factors of interest are adjusted within the pre-determined ranges with the unrelated factors strictly controlled. Those experiments often recruit a small number of healthy participants who will experience an adaptation period before the actual experiment (e.g., 30 min adaptation in [24,30,56], 55 min in [88], 60 min in [17, 25]). For most experiments, participants are tasked to perform

simulated office tasks, such as reading, listening, or typing (e.g. [55,88,90]). Other types of activities commonly observed in a non-working environment, such as eating, cooking, sleeping, or commuting, are less examined [25,47]. For experiments with multiple sessions, participants are exposed to short-term IEQ stimuli in each session, lasting from 20 to 180 min (e.g. [18,25,53,56,90,96]). Thus, their perceptions are directly linked to the IEQ stimuli in the experiment.

In contrast to climate chamber experiments, all the reviewed studies, except [25], developed overall satisfaction models in field evaluations. In those studies, the number of participants is much larger than in climate chamber experiments. The participants consist of different ages, genders, occupation groups, and their clothing insulations are often unrestricted. Moreover, field evaluations are generally longitudinal investigations in real-world buildings [40,54,68]. The built environments are maintained according to standards or guidelines but can vary with respect to outdoor climates and seasons, as opposed to the well-defined conditions in climate chambers. As a result, occupants' IEQ perceptions can be affected by factors that are not captured by the models. Another distinction between climate chamber experiments and field evaluations is that the latter often requires participants to provide feedback (either once or repetitively) about their general IEQ perceptions or past experiences (e.g. [62,63]), which may not directly reflect the conditions when the survey is being taken. Participation bias may also arise in field evaluations as dissatisfied occupants are more prone to provide feedback than those who are satisfied or indifferent [63,73]. In addition, researchers also face other challenges in field evaluations, including lack of building access and IECs [44,156], difficulties in recruiting participants, and cumbersome data collection [62].

Current models of overall satisfaction and IEQ interactions mainly focused on people aged between 17 and 50. Models of the very young and elderly populations are lacking, although those groups stand to benefit the most from such models. Besides the age groups, activities examined in climate chamber studies only represent the specific building types researchers aim to replicate (e.g., offices, classrooms). Other contexts, such as residential buildings, commercial buildings, and hospitals, are less frequently examined. Moreover, task performance measured in climate chambers is hardly the same as in the field evaluations [157], which can be due to 1) the Hawthorne effect [16]; 2) the different distraction levels in two scenarios; and 3) the use of proxy tasks rather than realistic tasks in climate chambers (e.g. [55,90]).

7.3. Climate variabilities

Occupants' IEQ perceptions and preferences are different across climate regions. Thermal comfort tolerance is largely climate-based due to the varying air temperature [14]. To achieve thermal comfort, buildings in the tropical climate in Malaysia use HVAC systems to maintain indoor temperature at around 24 °C–26 °C [158]. Cao et al. [159] indicated that the acceptable temperature range in winter is significantly different across climate zones in China: 20 °C–26 °C in severe cold zone, and 16 °C–22 °C in hot summer and cold winter zone. Northerners in China were also found to have higher demands for ventilation and natural lighting than southerners [32]. Besides, studies found that residents in Hong Kong [76] ranked the IAQ as the least important IEQ domain while Danes [160] and Swedes [161] considered it as the most important one. Thus, IEQ interactions and overall satisfaction models developed for a specific climate region can show low accuracy when applied to a different region.

7.4. Building types

Most studies reviewed in this paper are about offices (34 papers) and classrooms (13). A few studies investigate residential buildings (3), hospitals (2), shopping centers (2), hotels (2), and aircraft (1). Building types affect occupants' IEQ perceptions and expectations due to the varying functions they support, building designs, sizes, occupancy

levels, and environmental control systems. Thus, IEQ models developed for a particular building type can be invalid for others [69]. For example, Sun et al. [24] reported that occupants in offices preferred a higher illuminance and lower SPL than those in residential buildings. Thermal comfort is the most concerned IEQ domain in air terminals due to their unique architectural and operational features [84,162]. IEQ perceptions are also closely related to building designs. Open-plan offices can accommodate more occupants and facilitate team communication, when compared to private offices. However, they may reduce occupants' overall satisfaction due to small-sized workspaces, uncontrolled interactions, background noise, poor visual and speech privacy, if managed improperly [163,164]. Moreover, appropriate management of radiant asymmetry in buildings and adoption of biophilic design also have positive effects on occupants' thermal, visual and IAQ comfort.

7.5. Occupant data collection

Occupant survey is a common approach for occupant data collection. Paper-based survey is non-continuous, expensive, labor-intensive, and may induce participation bias. Although web-based and polling-based survey have been proposed to reduce the effort of data collection, they still suffer from low response rates, lack of long-term engagement, and survey fatigue [62]. Cost-effective, non-intrusive, and induced interactive feedback approaches for real-time and continuous data collection are thus highly desirable. For example, wearable sensors with feedback mechanisms [73], targeted right-now surveys with tracking systems [62], and intelligent IEQ monitoring and feedback system [165] have been explored in existing literature.

Data sufficiency and quality is another critical consideration in model development. Priori power analysis is utilized to determine the sample size for statistically significant results [18,24,56]. Critical analysis can select questions that provide useful information about comfort [54,166]. Techniques such as transfer learning [167] can help address the insufficient data by adapting the pre-trained models to a relevant target model using the new dataset. To measure occupant feedback, most studies used the ASHRAE seven-point Likert scales or the five-point variant. However, ambiguity and uncertainty may exist when occupants use those scales [61]. Moreover, data cleaning and processing techniques such as data normalization [24] and under-/over-sampling are often applied to manipulate the data in order to produce meaningful results [168].

7.6. Selection of model input features and output parameters

Selection of input features in analyzing interaction and combined effects is a major driver of inconsistent findings and hinders direct model comparison. In general, air temperature, relative humidity, mean radiant temperature, air velocity, and wet-bulb temperature are most frequently used to evaluate thermal comfort. Sound pressure level, sound types, and exposure time are common parameters for acoustic comfort. The input features for visual comfort include natural and artificial lighting, which are further measured by illuminance level, glare, and light color. For IAQ, researchers often choose ambient CO₂ concentration, VOC, PM_{2.5}, and odor. The suggested ranges of these parameters vary across seasons, building types, and climate zones.

There is no existing guideline or conclusion regarding which input features should be considered to quantify IEQ interactions and overall satisfaction. For example, some studies [57,64] used temperature to represent thermal comfort in the overall satisfaction model, while others [54,74] used the PMV instead. For interactions, temperature and SPL are frequently adopted to describe thermal and acoustic comfort [40,57,59], while other studies considered sound type as well [30,56,60]. Besides environmental features, input features from occupants also vary due to inconsistent survey design, the number of questions asked, measurement scales, and individual differences. Such inconsistencies also exist in model outputs when interpreting overall satisfaction (See Table 2).

Mihai and Iordache [58] used numeric ranges to indicate various satisfaction levels, while other studies adopted 0 or 0.5 as critical values or three-point scales (e.g. [54,55,64,75,169]).

7.7. Statistical analysis and modeling approaches

Linear regression has been the most popular approach to interpreting the relationships between IEQ and overall satisfaction [55,58,64,68,74]. Non-linear methods, such as logistic regression [75,76], geometric mean model [25,69], and harmonic mean model [25], have also been applied and showed better accuracy. ML algorithms, such as random forest [73,140,142,170], extreme gradient boost [140,170], ANN [170,171,172,173], classification trees [140,170], support vector machine [140,170], were initially used in thermal comfort prediction and have gained attention in predicting overall satisfaction [24,69]. Tang et al. [69] found that ML models (random forest and extreme gradient boost) showed better accuracy than regressions when working with a large number of IEQ domains and their complex interactions. However, the authors also suggested that those models were less generalizable, when compared to regressions. Moreover, interpretability is another challenge for ML models due to model complexity [174]. Other modeling approaches in existing research also include the open IEQ acceptance model [80], Bayesian IEQ acceptance model [71], AHP [61,81], Kano's model of satisfaction [17,82], and ensemble averaging method [85].

For IEQ interactions, the selection of statistical analysis and modeling approaches is based on the experimental design with the aim to interpret interactions between IEQ and their underlying environmental factors. ANOVA and *t*-test are most frequently used to examine the differences in IEQ perceptions between- and within-subjects across different experimental conditions [30,56,88,90,93,95]. Correlation analyses, such as Spearman's rank-order correlation [134], Pearson correlation [53,84], Kendall's correlation [28], and partial correlation [25], have been applied to describe the association between multiple IEQ domains. When an interaction is significant, researchers use the post-hoc test [30,89,90] to identify the causes of underlying differences. For non-normally distributed data, Wilcoxon matched-pair test [94,96] is used to compare the means of different conditions. Other statistical methods also play a similar role as ANOVA, including but not limited to the response surface methodology [22] and Newman-Keuls test [94,132]. To quantify the effect of interactions on IEQ comfort, the frequently adopted modeling approaches are linear mixed model [25,90,99], logistic mixed model [28], quadratic regression orthogonal model [22], and ANN [24].

8. Conclusion

Existing literature has extensively investigated single IEQ domains and their impacts on occupant satisfaction and performance. While there exist studies that examined multi-domain IEQ, they typically assess IEQ domains in isolation or consider two IEQ interactions at best. This paper conducted a comprehensive review of the interaction and combined effects of four IEQ domains, including thermal comfort, acoustic comfort, visual comfort, and IAQ. A broad, top-down approach with a three-stage screening process was adopted to select publications. In total, 74 publications were selected for a detailed review. In summary, four conclusions have been made:

- 1) The results of interaction and combined effects across studies are inconsistent. Possible causes include inconsistent generalized models, the varying experimental designs, climate variabilities, building types, data collection methods, selections of model input features and output parameters, and statistical analysis modeling approaches.
- 2) Among IEQ interactions, thermal and visual, thermal and acoustic, thermal and IAQ, and acoustic and visual interactions have been extensively studied while interactions of acoustic and IAQ, and

visual and IAQ are less explored. Three IEQ interactions are much less studied. Studies on four IEQ interactions are not found, possibly due to the complex factorial experimental design.

- 3) Current overall satisfaction models are generalized models which are trained on a group of occupants, and thus may fail to accurately predict an individual's satisfaction. Inspired by personal thermal comfort models, a personal overall satisfaction model involving multiple IEQ domains and their interactions is desirable in future research. Compared to linear approaches (e.g. [64,68]), recent literature has focused more on non-linear (e.g. [25,54]) and ML approaches (e.g. [24,69]) since they demonstrated better performance in interpreting the relationships between IEQ and overall satisfaction.
- 4) Independent of the modeling approaches, the one-vote veto effect has been observed in several studies [55,57,69,86]. This effect suggests that the relationship between IEQ and overall satisfaction is not a simple linear relationship in which each IEQ is associated with a constant weight. Instead, it emphasizes that dissatisfaction with a single IEQ domain can predominate the overall satisfaction regardless of the performance of other IEQ.

Given the current small-scaled, fragmented, and inconsistent findings on the IEQ interaction and combined effects, an international collaboration on developing a global multi-domain IEQ database is a worthwhile effort to promote future research and best practices. In this open database, occupant "right-here-right-now" survey and environmental data can be shared with other information, such as experiment dates, participant characteristics, accessibility to IECs, building types, building codes, geographic locations, seasons, and outdoor climates. A user-friendly webpage can enable users with different software skills to easily access, navigate, and download data from this database and contribute to it. This database will advance theoretical and practical knowledge of IEQ in two ways. First, researchers and practitioners can generate new IEQ or overall satisfaction models, benchmark model performance, and compare results in different contexts (e.g., building types, occupants). Second, this database is also expected to guide the future improvement of building standards (e.g., ASHRAE Guideline 10 [175]) which highlights current evidence of IEQ interactions is limited [72]. Such standards will further enable building operators to make informed decisions about building management, such as how to address the worst IEQ to improve overall satisfaction efficiently.

CRedit authorship contribution statement

Yijin Zhao: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation. **Da Li:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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